

A Finite Monotone Axial Clock for the Quartet Shape Flow

Sixteen Scales, Two Channels, and a Narrow Asymmetry Corridor

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Abstract

RH-213 reduces each centered physical quartet to exact shape coordinates $(u, \eta) \in [0, 1] \times [-1, 1]$. We now audit their scale ordering on sixteen small-noise levels from 0.04 to 0.00125. On both physical channels, u increases strictly through all fifteen coarse-to-fine transitions. The net change is 0.69809 on the left and 0.69673 on the right; the smallest observed increments are 0.01529 and 0.01513.

After the initial transient, defined in advance here by $\sigma \leq 0.02$, η remains in narrow corridors: $[-0.10444, -0.06886]$ on the left and $[-0.10748, -0.07031]$ on the right. Their widths are 0.03557 and 0.03717. Mature left/right discrepancies are at most 0.003006 in u and 0.003043 in η .

Any strictly ordered finite data define an invertible piecewise-linear clock on their sampled log-scale interval; we state this elementary theorem to separate a legitimate finite clock from an unsupported all-level law. The observed trajectory is therefore well described as one dominant axial motion plus a bounded transverse corridor, but no monotonicity theorem below the finest level, autonomous flow, or small-noise limit is claimed.

1 Exact coordinates and finite question

For each physical channel $s \in \{L, R\}$ and scale σ , RH-213 writes the centered-RMS quartet as

$$\sqrt{u_{\sigma,s}} \pm i\sqrt{(1 - u_{\sigma,s})(1 + \eta_{\sigma,s})}, \quad -\sqrt{u_{\sigma,s}} \pm i\sqrt{(1 - u_{\sigma,s})(1 - \eta_{\sigma,s})} \quad (1)$$

[2]. The coefficient vector is an exact function of these two coordinates. RH-212 supplies a frozen sixteen-level root atlas [1].

We ask three finite questions:

1. Is u ordered consistently as σ decreases?
2. Does η enter a narrower post-transient corridor?
3. Do both physical channels track the same coordinate path within small finite discrepancy?

None of these questions presupposes a limiting value.

2 Scale order and clock convention

Write

$$t = \log(1/\sigma). \quad (2)$$

The sixteen anchors are ordered by increasing t , equivalently decreasing σ . Let $t_0 < \dots < t_{15}$ and let $u_k = u_{\sigma_k, s}$ on one channel.

Theorem 2.1 (Finite monotone-clock interpolation). *If*

$$u_0 < u_1 < \dots < u_n, \quad (3)$$

then there is a unique continuous function $U : [t_0, t_n] \rightarrow [u_0, u_n]$ that is affine on each interval $[t_k, t_{k+1}]$ and satisfies $U(t_k) = u_k$. It is strictly increasing and hence a homeomorphism onto its image.

Proof. Linear interpolation is unique on every subinterval. Its slope there is $(u_{k+1} - u_k)/(t_{k+1} - t_k) > 0$, so the pieces join to a strictly increasing continuous bijection. A continuous strictly monotone map on a compact interval has continuous inverse. $\square$

The theorem is intentionally finite. It neither extends U beyond the sampled interval nor says that the piecewise-linear interpolation is the physical law.

3 Sixteen-level axial verdict

For each transition define

$$\Delta u_{k,s} = u_{k+1,s} - u_{k,s}. \quad (4)$$

All thirty channel-transition increments are positive.

axial diagnostic	left	right
transition count	15	15
positive increments	15	15
minimum increment	0.015295	0.015130
maximum increment	0.104022	0.106002
net change	0.698086	0.696733
total variation	0.698086	0.696733
rank correlation with $\log(1/\sigma)$	1	1

Equality of total variation and net change is simply another expression of finite monotonicity. The factor-of-seven spread between minimum and maximum increments warns against treating the anchor index itself as constant-speed time.

4 Representative dyadic subsequence

The following equal-log-step subsequence displays the dominant motion:

σ	u_L	u_R	η_L	η_R
0.04	0.02027	0.02077	-0.57806	-0.59358
0.02	0.24887	0.24794	-0.06886	-0.07031
0.01	0.40529	0.40229	-0.10443	-0.10748
0.005	0.54098	0.54034	-0.08934	-0.09080
0.0025	0.65874	0.65801	-0.09050	-0.09181
0.00125	0.71835	0.71750	-0.09303	-0.09444

The first transition contains a large transverse relaxation. Thereafter the axial coordinate advances while η fluctuates around roughly -0.09 .

5 The mature asymmetry corridor

For a finite ordered sequence x_k , define

$$\text{width}(x) = \max_k x_k - \min_k x_k, \quad \text{TV}(x) = \sum_k |x_{k+1} - x_k|. \quad (5)$$

On the levels with $\sigma \leq 0.02$:

transverse diagnostic	left	right
minimum η	-0.104434	-0.107477
maximum η	-0.068864	-0.070311
width	0.035570	0.037166
mean	-0.089618	-0.091187
RMS about mean	0.009378	0.009636
total variation	0.070188	0.074222

The total variation is about twice the width, so η is not monotone. A “transverse fixed point” would overstate the evidence. The justified term is a finite corridor.

6 Dual-channel synchronization

At fixed σ , define

$$\delta_u(\sigma) = |u_{\sigma,L} - u_{\sigma,R}|, \quad \delta_\eta(\sigma) = |\eta_{\sigma,L} - \eta_{\sigma,R}|. \quad (6)$$

Across all sixteen scales,

$$\max \delta_u = 0.004229, \quad \max \delta_\eta = 0.015523, \quad (7)$$

where the transverse maximum occurs in the initial transient. On the mature subset,

$$\max \delta_u = 0.003006, \quad \max \delta_\eta = 0.003043. \quad (8)$$

These discrepancies are small compared with the net axial displacement, but they are not interval-certified errors. They quantify agreement of two finite discretizations/channels under the same root-selection rule.

7 A near-one-dimensional description

The phrase “near one-dimensional” has a precise finite meaning here:

1. one coordinate is strictly ordered through every observed transition;
2. the other remains in a post-transient interval of width below 0.038;
3. both channels agree to roughly 3×10^{-3} in each mature coordinate.

It does not mean that η is a function of u at all levels, that the trajectory lies on an invariant curve, or that a one-dimensional generator has been found.

8 Why finite monotonicity matters

The raw quartic coefficient flow looked irregular because location, radius, axial separation, and transverse asymmetry were mixed. The exact shape coordinates show that one component has a stable order. This gives a useful clock for:

1. predeclared out-of-sample extrapolation;
2. boundary-distance estimates;
3. sensitivity decompositions;
4. equal-log-step recurrence audits.

These uses remain finite unless accompanied by uniform estimates.

9 Claim boundary and next test

The data do not prove:

- u_σ is monotone for every sufficiently small σ ;
- u_σ has a limit, much less limit one;
- η_σ converges;
- a scale-independent map advances the shape;
- a growing determinant exists.

RH-215 therefore freezes seven fine levels as training data and withholds two finer levels for prediction. Gate A stays open, and no statement about Gates B–E or zeta zeros is made.

References

- [1] Liang Wang. An intrinsic normalization audit for the physical quartic divisor, 2026. RH-212 preprint.
- [2] Liang Wang. The centered conjugate-quartet shape manifold, 2026. RH-213 preprint.